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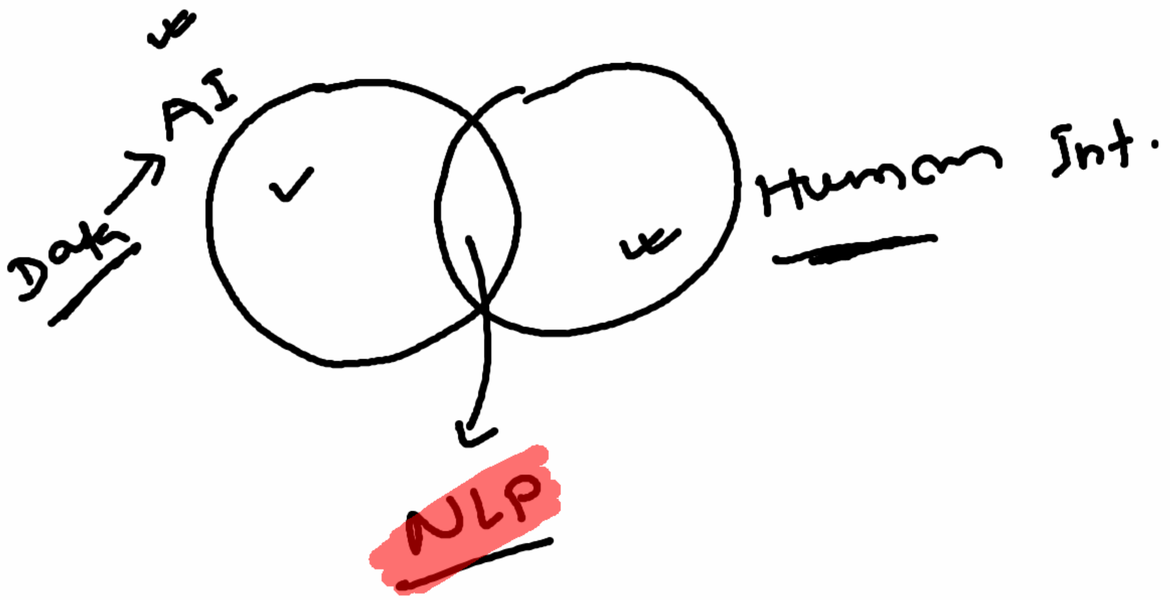

Talking: Learnvista Private Lim...

1956
Rosenblatt - Perceptron
AI term
USA - AI
Yann Lecun
Russian In
LLM
GANS
ACH
MLP
1990
1986
2006
Geoffrey Hinton
GPU
2012
AI Winter
Modern Deepl

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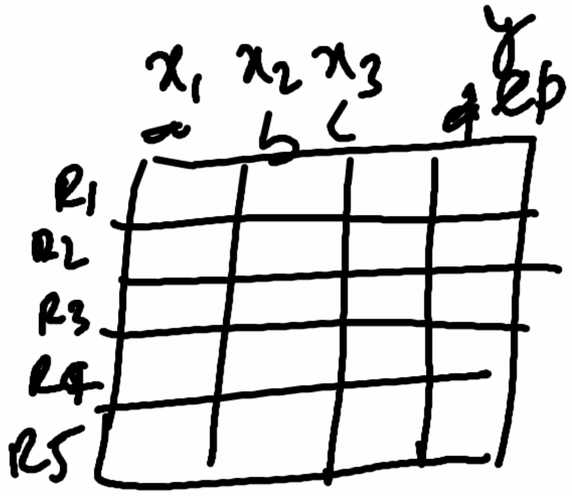

Algorithms

Tabular data → Supervised ML / MLP

Channel

RGB
Gray

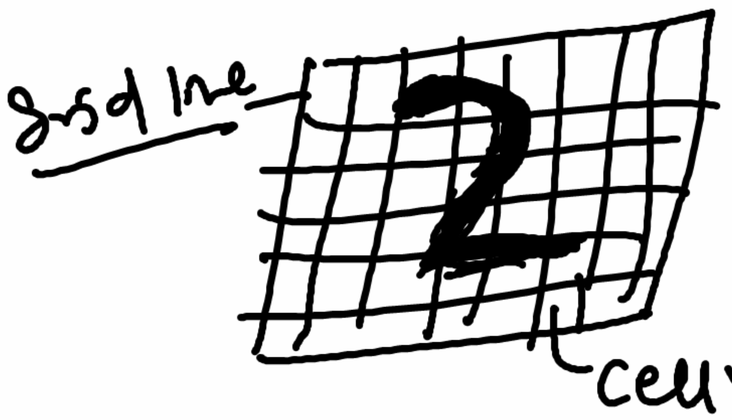

image
Pixel
cell value

Pixel size

Hi, my name is Sundram

Algorithms

sequence

is Hi Sundram naming

Algorithms

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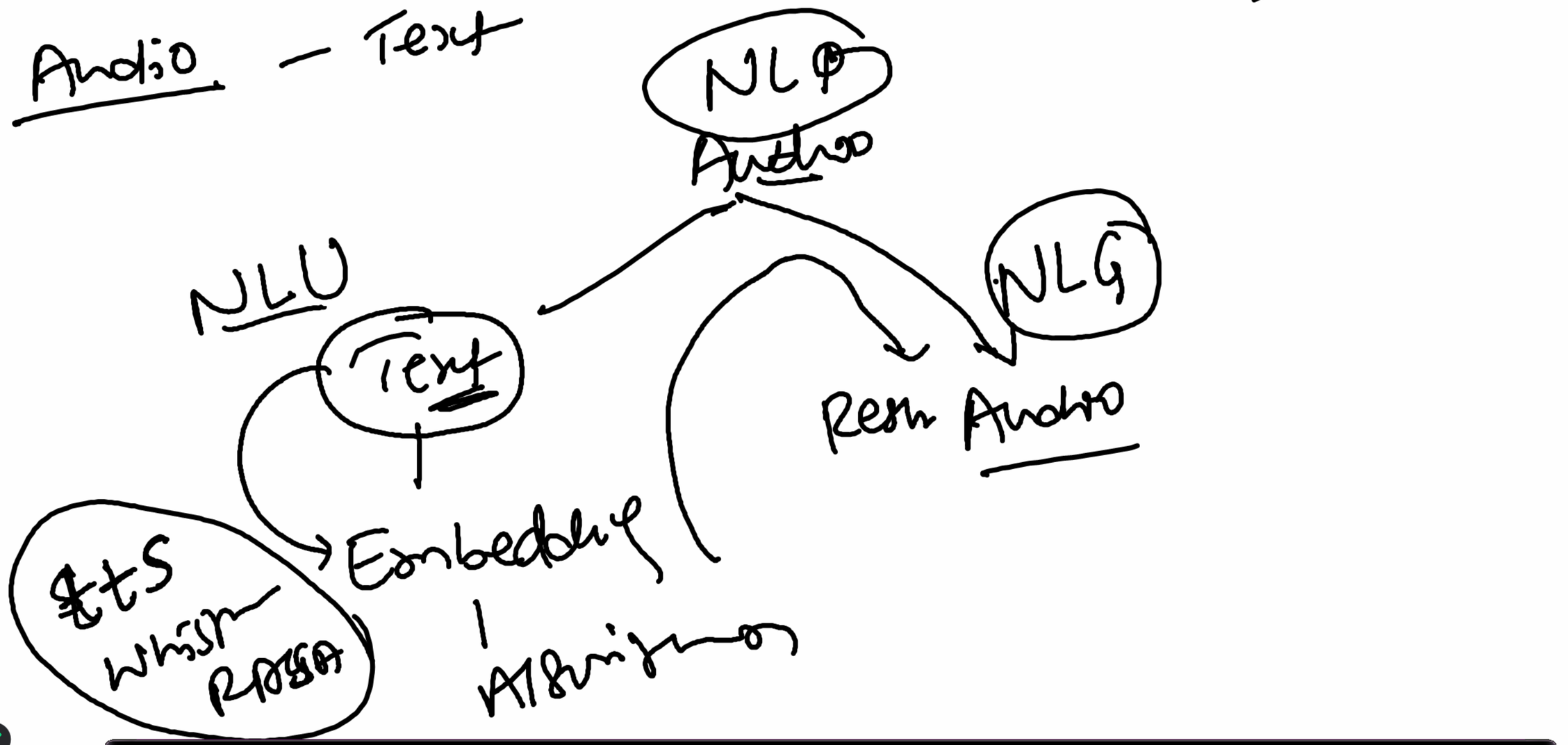
Talking:

LLM - max^on 801 → Text only

Audio - Text



Audio Video Participants 36 Chat 2 Share Pause Layout Annotate Remote control Show meeting More
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NLP

ML

Basic Language model

- Bag of words
- TF-IDF
- N-Grams

Text to vector

Tokenization
Stemming / Lemmatization
Stopwords etc

Distributed Language model

- Word2vec – Google
- Glove – Stanford
- Fast-Text – Facebook/Meta

Text to vec

Algorithm – Deep learning
RNN, LSTM, GRU
Seq-2-Seq

Gen AI

Large Language model (LLM)

Embedding
- Transformers – 2017
- BERT – 2018
- GPT – 2018

Fine-tuning PEFT
LLM model

I/P
Adapter
LORA
Q-LORA

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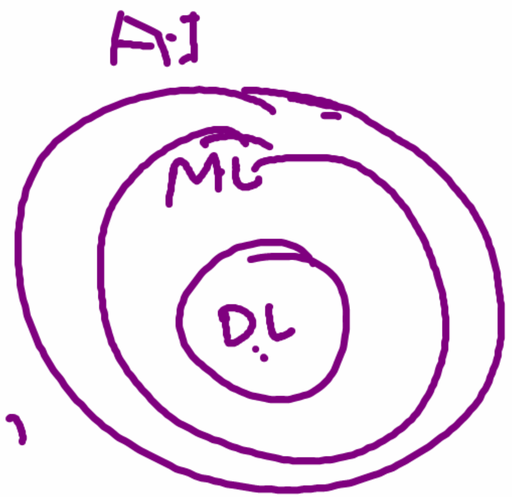
ML

- → manual work
- → manual feature handling
- → Hardware - CPU
- → Software - Sklearn, numpy, Pandas, matplotlib, seaborn etc
- → Your home Comp
  - ✓ ↳ Interpretability

DL

- → Auto matic
- → Automatic - feat
  - ↳ Hidden Layer
- → Hardware - GPU
- → Software - Tensorflow, Keras, Pytorch, Theano, NLTK, Spacy, Gensim - -
- → NO Comp —
  - ↳ Yes —



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R1 → I like Food. [1110000000 . . . 0]
R2 → I do not like food
R3 → Food was simply awesome
R4 → One of the pathetic food I ever ate.
R5 → Food was delicious and affordable.

} Sentiment / Text

Vocab - unique = 17 (sequence)

BOW < Binary BOW / Count BOW → Sparse matrix

	I	like	Food	do	not	was	simply	awesome	one	of	the	pathetic	ever	ate	deli	and	affo
R1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0
R2	1	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0
R3	0	0	1	0	0	1	1	1	0	0	0	0	0	0	0	0	0
R4	1	0	1	0	0	0	0	0	1	1	1	1	1	1	0	0	0
R5	0	0	1	0	0	1	0	0	0	0	0	0	0	0	1	1	1

Bag of words (BOW) — Binary BOW

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R1 - Food is good = (+)ve

R2 - Food is not good = (-)ve

$R1 \neq R2$

Stopwords —

R1 - Food good

R2 - Food good

$R_1 = R_2$

BOW

Uni-gram - 1

(n)-grams — Bi-gram - 2

tri-gram - 3

tetra - 4

$R1 = [1\ 1\ 0]$

$R2 = [1\ 0\ 1]$

$R_1 \neq R_2$

Vocab :-

Bi-gram

	Food is	is good	is not	not good
R1	1	1	0	0
R2	1	0	1	1

$R_1 \neq R_2$